MILITARY SPECIFICATION

INSULATION SHEETS, ELECTRICAL, LIGNO-CELLULOSIC, HIGH DENSITY, HARDBOARD

1. SCOPE

1.1 This specification covers fiberistic, ligno-cellulosic high density, laminated sheet materials.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
PPP-B-636 - Box, Fiberboard.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

MILITARY
MIL-B-43291 - Box, Fiberboard, Corrugated, Double-Wall, Weather Resistant

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129 - Marking For Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.
National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D-229 - Testing Rigid Sheet and Plate Materials Used For Electronic Insulation.
D-635 - Test For Flammability of Rigid Plastics Over 0.127 cm. (0.050 in.) In Thickness.
D-952 - Test For Bond Strength of Plastics and Electrical Insulating Materials.
D-1037 - Evaluating the Properties of Wood-Base Fiber and Particle Panel Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.2 (see 6.3).

3.2 Material.

3.2.1 Construction. The material shall consist of interfelted, ligno-cellulosic fibers consolidated under heat and pressure. Two or more individual sheets may be laminated to the desired thickness.

3.2.2 Property values. The material shall conform to the property values shown in table III. The values obtained for each set of specimens taken from the same sheet shall be averaged before comparison with the applicable table.

3.2.3 Conditioning of samples. Samples for first article and conformance testing shall be conditioned, as described in 3.2.3.1 through 3.2.3.4 for the tests listed in tables III and IV.

3.2.3.1 Samples shall be tested in the condition received.

3.2.3.2 Samples shall be subjected to conditions as outlined in the applicable test procedure.

3.2.3.3 Samples shall be subjected to 90° Fahrenheit - 90 percent relative humidity for 96 hours prior to testing. Surfaces shall be wiped dry and tests commenced within 30 minutes after removal from conditioning environment.

3.2.3.4 Samples shall be subjected to a 90° Fahrenheit - 90 percent relative humidity for 96 hours prior to testing. Surfaces shall be wiped dry and tests commenced within 30 minutes after removal from conditioning environment.

3.2.4 Uniformity. Both surfaces of all sheets of any lot shall be smooth, uniform in texture, finish and color.

3.2.5 Surface defects. The material shall be free from internal and external blisters delamination or cracks, and reasonably free from other small defects such as scratches, dents, or heat marks.

3.2.6 Warp or twist. (Applicable only to uncut sheets as received from manufacturer.) The warp or twist of material as delivered shall not exceed the values of table I.

Table I - Warp or twist.

Thickness Inch	Maximum permissible warp or Twist 1/ (on basis of 36 inch dimension) Percentage
1/8 to 1/4	1.0
3/8 to 3/4	0.5
Over 3/4	0.25

1/ In case of warp, this percentage is stated in terms of the lateral dimensions (length or width), in the case of twist, the percentage is stated in terms of the dimension from one corner to the diagonally opposite corner.

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3.3 Dimensions and tolerances.

3.3.1 Length and width. Unless otherwise specified by the procuring activity, the manufacturer's standard sizes of 48 inches in width and between 36 and 144 inches in length will be acceptable. The length and width of sheets may vary 1 inch over or under the manufacturer's standard size. Non-uniform standard sheet dimensions caused by cutting specimens for tests required by this specification shall not be cause for rejection, unless particular sheet dimensions are specified (see 6.2).

3.3.2 Thickness. Sheets shall be furnished in the nominal thickness shown in table II, as specified (see 6.2). At least 90 percent of the area of the panel shall be within the tolerance range given. At no point shall the thickness vary more than one-eighth of the range below the minimum or above the maximum.

Table II - Thickness.

Nominal thickness Inches	Decimal equivalent in inches	Permissible variation	
		Minimum	Maximum
1/8	0.125	0.125	0.155
1/4	0.250	0.238	0.262
3/8	0.375	0.358	0.392
1/2	0.500	0.479	0.521
5/8	0.625	0.601	0.649
3/4	0.750	0.723	0.777
1	1.000	0.967	1.033
1-1/4	1.250	1.213	1.287
1-1/2	1.500	1.459	1.541
2	2.000	1.951	2.049

3.4 Color. The material shall be furnished natural in color which is the color naturally attained in the manufacturing process without the addition of any colorants.

3.5 Machineability. The material shall be such that it can be drilled, tapped, sawed, and machined in all directions [1/] without cracking, splitting, or otherwise impairing the material for general use.

3.6 Surface finish. Unless otherwise specified by the procuring activity, the surface finish of the sheets shall be semi-gloss on both faces as produced by the laminating process.

3.7 Workmanship. Laminated materials shall be manufactured and processed in a careful and workmanship manner, and shall be free from defects, other than with respect to color (see 3.4), which may affect the appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. First article inspection shall consist of the examination and tests specified in table III. First article tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center, Materials Development and Applications Office, Department of the Navy, Washington, D. C. 20360.

[1/] The nature of laminated material necessitates special precautions when drilling and tapping parallel to laminations.

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Table III - Tests for property values.

	CONDITIONING	TEST PROCEDURE	UNIT OF VALUE	THICKNESS (Inches) 1/8	1/2
ELECTRICAL PROPERTY TO BE TESTED					
Dielectric Strength (Step-by-Step) in Oil - Parallel to Laminations					
As Received	3.2.3.1	ASTM D-229 Tapered pin method	KV. Min. avg	30	30
Conditioned	3.2.3.3	ASTM D-229 Tapered pin method	KV. Min. avg.	30	30
Arc Resistance	3.2.3.2	ASTM D-229	Seconds, Min.	55	55
Insulation Resistance	3.2.3.4	ASTM D-229 Tapered pin method	Megohms, Min. avg.	250	250
Dielectric Constant					
60 Cycle	3.2.3.2	ASTM D-229	Min. avg.	7.0	7.0
1 Megacycle	3.2.3.2	ASTM D-229	Min. avg	5.5	5.5
PHYSICAL PROPERTY TO BE TESTED					
Specific Gravity	3.2.3.2	ASTM D-1037	Minimum	1.15	1.28
			Maximum	1.30	1.43
Flexural Strength	3.2.3.2	ASTM D-229	Psi - Min.	10,000	9,000
Modulus of Elasticity	3.2.3.2	ASTM D-1037	Psi - Min	1,100,000	1,100,000
Compressive Strength	3.2.3.2	ASTM D-229	Psi - Min.	21,000	21,000
Bond Strength	3.2.3.2	ASTM D-952	Psi - Min.	300	300
Surface Hardness	3.2.3.2	ASTM D-229	Rockwell -M Scale - Min.	80	80
Water Absorption-24 Hr.	3.2.3.2	ASTM D-1037	Percent by Weight - Max.	8.0	2.0
Modulus of Rupture	3.2.3.2	ASTM D-1037	Psi - Min.	10,000	9,000
Flammability	3.2.3.2	ASTM D-635	Pass or Failure	Self extinguishing and non-burning	

4.3.2.1.1 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be one sheet.

Examine	Defect
Appearance	Color, not as specified (see 3.4) Non-uniformity of texture or finish (embossments). Surface, water spots, oil spots Presence of foreign material in surface of imbedded particles.
Workmanship	Delamination of plies. (Edge checking as a superficial defect is acceptable).

4.3.2.1.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be one sheet.

Examine	Defect
Length and width	Varies by more than tolerances indicated in 3.5.1 (commercial sizes).
Thickness	Varies by more than tolerances indicated in table II.

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4.3.2.1.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking comply with the requirements of section 5 of this specification. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation.

Examine	Defect
Packaging	Not level specified. Materials or construction not as specified.
Packing	Container material or construction not as specified. Any non-conforming component, component missing or otherwise damaged, affecting serviceability.
Count	Less than specified or indicated quantity of sheets.
Weight	Gross weight exceeds specified requirements.
Markings	Exterior markings omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.
Closure, waterproofing, and reinforcing	Not as specified.

4.3.2.1.4 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality levels (AQLs) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.2.1.1	I	1.5
4.3.2.1.2	I	2.5
4.3.2.1.3	I	4.0

4.4 End item testing.

4.4.1 Quality conformance testing. Quality conformance testing of the end item shall be conducted in accordance with Table IV for the characteristics as indicated herein, for each lot submitted.

Table IV - Quality conformance testing.

Characteristic	Conditioning	Test Procedure	Unit of value	Requirement
Caliper	3.2.3.1	3.3.2	Pass or Failure	Pass
Warp or Twist	3.2.3.1	3.2.6	Pass or Failure	Pass
Specific Gravity	3.2.3.1	ASTM D-1037		
1/8 in.			Minimum	1.15
			Maximum	1.30
1/4, 3/8, and 1/2 in.			Minimum	1.28
			Maximum	1.43
3/4 in. and Over			Minimum	1.35
			Maximum	1.50
Modulus of Rupture	3.2.3.1	ASTM D-1037		
1/8 in.			Psi Minimum	10,000
1/4 in. and Over			Psi Minimum	9,000
Bond Strength	3.2.3.1	ASTM D-952	Minimum	350
Water Absorption-24 Hr.	3.2.3.1	ASTM D-1037		
1/8 in.			Percent Max.	8.0
1/4 and 3/8 in.			Percent Max.	2.5
1/2 in			Percent Max.	2.0
5/8 and 3/4 in.			Percent Max.	1.5
1 in. and Over			Percent Max.	1.0

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Domestic shipment and immediate use.

5.1.1 Packaging. Packaging shall be sufficient to afford protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier, at the lowest rate, and will afford protection against physical damage during direct shipment from the supply source to the first receiving activity for immediate use. The shipping containers or method of packing shall conform to the Uniform Freight Classification or National Motor Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.3 Marking. Shipment marking information shall be in accordance with MIL-STD-129.

5.2 Domestic shipment and storage or overseas shipment. The levels of packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.2).

5.2.1 Packaging.

5.2.1.1 Level A. Sheets shall be wrapped individually, or interleaved with a non-bleeding, barrier material conforming to PPP-B-1055.

5.2.1.2 Level C. Packaging shall be sufficient to accord adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use.

5.2.2 Packing.

5.2.2.1 Level A. Sheets, packaged as specified (see 6.2), shall be packed in containers conforming to class weather-resistant of PPP-B-636 or MIL-B-43291. All center and edge seams and manufacturer's joints of the container shall be sealed and waterproofed with pressure sensitive tape and container reinforcing shall be accomplished by use of reinforced, pressure-sensitive tape in accordance with the appendix to the applicable box specification. Container gross weight shall not exceed the weight limitations of the applicable box specifications.

5.2.2.2 Level B. Sheets, packaged as specified (see 6.2), shall be packed in containers conforming to class domestic of PPP-B-636. Shipping containers shall be closed in accordance with the box specification or appendix thereto. Fiberboard boxes shall not exceed the weight limitations of PPP-B-636.

5.2.2.3 Level C. Sheets, packaged as specified (see 6.2), shall be packed in containers, at the lowest rates, in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first receiving activity for immediate use. This level is general shall conform to the Uniform Freight Classification Rules, National Motor Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.2.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The insulation sheet material covered by this specification is intended primarily for use as electrical and electronic workbench tops.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sheet size and thickness required (see 3.5).
(c) Whether sheets, smaller in dimension than specified in the procurement document due to cutting specimens for lot acceptance tests are acceptable (see 5.3.1).
(d) Levels of packaging and packing required (see 5.1 and 5.2).
(e) Special marking required (see 5.1.3 and 5.2.3).

6.3 First article

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 5970-N0271)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

DD FORM 1 APR 53 1426 REPLACES NAVSHIPS FORM 4883, WHICH IS OBSOLETE (NAVSHIPS 0VPRT 12-86)

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